

PROBLEM SET 1

COM SCI 188: Intro to Robotics

Spring 2025

Question:	1	2	3	4	5	6	Total
Points:	10	10	16	18	22	24	100

1. (10 points) **Definition of Robots.** Are the following a robot? Explain why.

(a) ChatGPT:

(b) Washing machine:

(c) Remote-Controlled car:

(d) SpaceX rocket booster catcher:

(e) Amazon Echo device with Alexa:

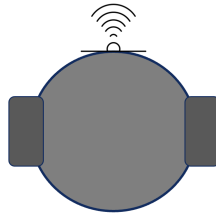
2. **Sensor Types.** Robots use a variety of sensors to understand their internal states and interact with their environments.

(a) (4 points) Define and distinguish between active versus passive sensors, and proprioceptive versus exteroceptive sensors.

(b) (6 points) Classify the following sensors accordingly (justify accordingly):

- GPS:
- Bumper Switch:
- Accelerometer:
- LIDAR:
- Infrared proximity sensor:
- Wheel encoder:

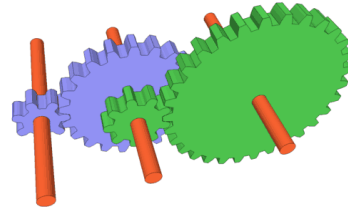
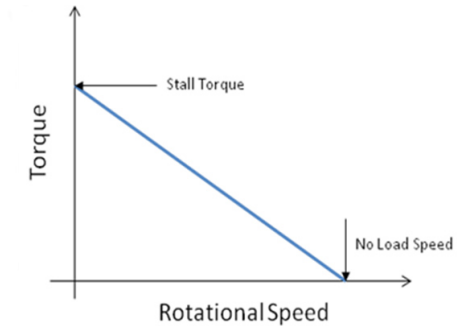
3. **Sonar.** Assume you have a single fixed sonar sensor (with a angular range of 30 degrees) on your two-wheel robot as shown below:



- (a) (4 points) Explain why having only a single fixed sonar sensor is unlikely to be enough for obstacle avoidance.
- (b) (4 points) You decide to add more sonars to this robot. Due to the environment and task nature, you only care about detecting obstacles as the robot drives forward. What is the minimum number of (fixed) sonars needed to detect all obstacles in this case?
- (c) (8 points) Which of the following are limitations of sonar sensors in indoor environments?
- A. Inaccurate readings on soft or angled surfaces
 - B. Requires visible light to function
 - C. Confusing echoes in enclosed spaces
 - D. Low angular resolution (difficulty detecting narrow objects)
 - E. Strong interference from magnetic fields
 - F. Susceptible to ambient sound and ultrasonic noise
 - G. Poor performance in low-light conditions
 - H. Difficulty measuring distance to very small objects

4. **Motor & Gears.** Suppose you find a DC motor sitting around the lab and need to characterize it before using it in your wheeled mobile robot. Assume you have all the tools required to take the necessary measurements.

Hints:

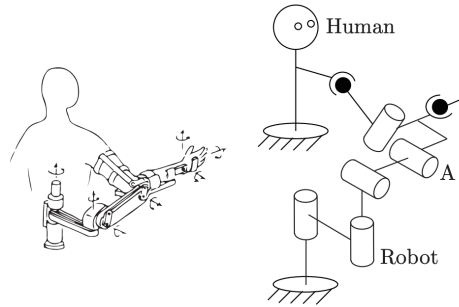


- (a) (6 points) You turn on the motor and find that its speed without any attached load is 1000 RPM (revolutions per minute). When you apply a torque of 0.5 kg-cm to the motor, the speed decreases to 250 RPM. What is the stall torque of the motor?
- (b) (6 points) After finding the above stall torque, you determine that this motor is not strong enough for the robot you are building, so you decide to add a gearbox. You want a minimum stall torque of 8 kg-cm. Design a multi-stage gearbox to achieve this.
- (c) (6 points) After adding the gearbox, what is the maximum speed that your robot can travel at? Assume a wheel diameter of 4 cm.

5. Degrees of Freedom.

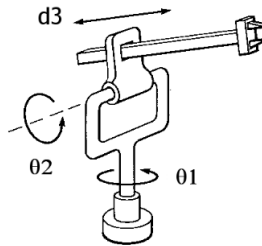
- (a) (4 points) Find the number of degrees of freedom of your arm, from your torso to your palm (just past the wrist, not including finger degrees of freedom). Keep the center of the ball-and-socket joint of your shoulder stationary (do not “hunch” your shoulders). Provide a brief justification.

- (b) (6 points) The figure below shows a robot used for human arm rehabilitation:



Determine the number of degrees of freedom of the chain formed by the human arm and the robot.

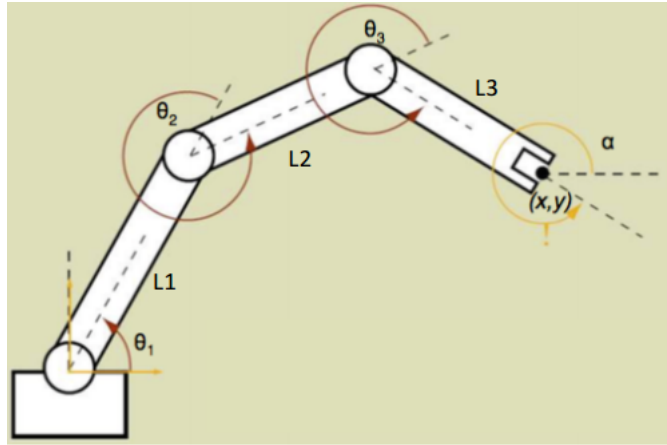
- (c) (12 points) Consider the following RRP manipulator:



Write down the Denavit-Hartenberg parameters for this manipulator, in the form of a table:

i	a_{i-1}	α_{i-1}	d_i	θ_i
1				
2				
3				

6. **Rigid Body Motion.** Planar manipulator robot pictured below. The robot has three DC motors attached at joints. (Note: the robot's end-effector only travels in the plane of the page.)



- (a) (4 points) What is the number of DoF of this arm?
- (b) (4 points) For a given reachable end-effector position in 2D space, how many inverse kinematic solutions can this robot theoretically have?
- (c) (6 points) Given $L_1 = 0.6$, $L_2 = 0.4$, $L_3 = 0.3$, $\theta_1 = 60^\circ$, $\theta_2 = 330^\circ$, $\theta_3 = 315^\circ$. Calculate the position (x, y) of the end-effector in the base frame.
- (d) (10 points) Explain the concept of a Jacobian matrix in the context of robotic manipulators. Then, derive and write the Jacobian matrix for this specific manipulator.